1 A Messy Literature Review

Fama and MacBeth, 1973, Ma et al., 2020. Fama French 3 and 5 factors. GMM 1982. Ghysel's critique of the dynamic CAPM. Also should look at the local linear and local maximum likelihood estimation methods. Jagannathan and Z. Wang, 1998b provides a good reading on the subject. Shanken, 1992a on the standard error of the Fama-Macbeth Regression.

For example, the CAPM assumes one factor that is the market excess return r_m.

$$E(r_i) = \beta_i E(r_m) \quad \text{where} \quad \beta_i := \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)}$$

Fama and MacBeth, 1973 derive testable restrictions from the CAPM model and embed the model in a linear regression:

$$r_{it} = \gamma_{0t} + \gamma_{1t}\beta_i + \gamma_{2t}\beta_i^2 + \gamma_{3t}s_i + \varepsilon_t$$

where β_i is estimated with $\hat{\beta}_i$ from time series regression for each i

$$r_{it} = \alpha + \beta_i r_{mt} + u_{it}$$

The testable hypotheses are

1. $E(\gamma_0) = 0$ there is no pricing error in the expected excess return.
2. $E(\gamma_2) = 0$ the linearity test.
3. $E(\gamma_3) = 0$ there is no other risk measure.
4. The Sharpe-Lintner Hypothesis that $E(\gamma_1) = R_{ft} > 0$.

Ma et al., 2020 considers Gibbons, Ross, and Shanken, 1989 tests with large N and points out that the factor loadings change substantially over time, they conduct a High-dimenisonal alpha test and

Jegadeesh et al., 2019 and Chaieb, Langlois, and O. Scaillet, 2018 are new papers that referenced the Jagannathan and Z. Wang, 1998b, see also Jagannathan and Z. Wang, 1998a.

Generally, people test CAPM or other asset pricing models by testing that the $\alpha = 0$ (Jensen's α), meaning there is no systematic pricing error(in conditional expectation).

K. Q. Wang, 2003 does a testing of CAPM and factor models in a nonparametric setting with conditioning information. The starting point is the SDF characterisation of excess returns r_i:

$$E_t(m_{t+1} r_{i,t+1}) = 0 \implies E_t\left(\left[1 - r_{m,t+1}\frac{E_t(r_i, r_m)}{E(r_m^2)}\right] r_{i,t+1}\right) = 0 \tag{1}$$

Then $m_{t+1} = 1 - b(x) r_{m,t+1}$ is nonparametrically estimated from the market excess return $r_{m,t}, t = 1, \dots, T$. The covariates x is four-dim in the empirical part. Then they test relationship by the following regression(they use weighted Least Square for technical reason).

$$\hat{m}_{t+1} r_{m,t+1} = z_t' \delta_i + u_{i,t+1} \quad \text{where} \quad z_t \in I_t$$

For testing a economic theory in a nonparametric setting, see Matzkin, 1994.

For *error-in-variable* problem, there is an Fan and Truong, 1993 on nonparametric regression with error in variables, they proposed deconvolution method to account for the observation errors. There is Bickel and Ritov, 1987 on efficient parametric regression: there is identification problem in the error-in-variable problem, which serves as an important example in Rothenberg, 1971. Reiersøl, 1950 is a paper on the identification issue, it shows equivalence with the assumption of normal measurement error and show equivalence using characteristic functions, this method should be noted.

For *cross-sectional correlation problem*, we can see Li and Racine, 2007 chapter 18 and also Robinson, 2012.

Acharya and Pedersen, 2005 proposes a liquidity factor and uses the cross-sectional method to verify. Further applications, see Jegadeesh et al., 2019 page 2. Jegadeesh et al., 2019 proposes an IV method to deal with the first step estimation error, or error in variable, this allows them to use individual stocks instead of portfolios, they do the test on CAPM **and** factor models.

Apparently Fama Macbeth didn't give statistical reasons for the two-pass regression procedure. Gibbons, Ross, and Shanken, 1989, Gibbons, 1982 and Shanken, 1992a should be necessary companion.

Shanken, 1992a mentioned that the analysis of MLE is not easy because they have

N, T both $\longrightarrow \infty$. The underlying model is richer than just simply APT, but see Shanken 1982 1985 1992 on APT and Dybvig and Ross, 1985 for discussion.

Shanken, 1992a gives a general parametric framework where he distinguished the factors that are and are not returns, and derive asymptotic distribution of the two-pass estimator of the γ's. The first lemma regards the first step estimation.

See Ross for APT. Dybvig and Ross, 1985 says that APT theory incoporates assumptions other than a simple factor generating structure so any test of APT should take account of that. Chamberlain, 1983 is also a must-read.

Herskovic, 2018 referneced the Shanken, 1992a.

There is a different paper Shanken, 1992b. For the Eubank and Spiegelman, 1990 might be useful for the inference. Hardle and Mammen, 1993.

Follow the line of specification test as in Hardle and Mammen, 1993, there is a paper Chen, Gao, and Tang, 2008 which gives an extensive review of relevant literature. Two of the papers are worth mentioning, the genralized likelihood method and the test of $\hat{\varepsilon} = 0$. But it requires some ordering of the observations, which seems a bit odd. The empirical likelihood method and generalized likelihood ratio test seems to be important in nonparametric inference literature, should check it out. Fan and Zhang, 2004 puts an even more general likelihood ratio where the error term is no longer required to be iin a parametric family.

There seems to be a lot of specificaton tests. and Fama-Macbeth is quite popular, Feng, Giglio, and Xiu, 2017 is an example, where they use double Lasso selection to both select the relevant factor and keep some of irrelevant factors in the regression to avoid the omitted variable problem and then they use Fama-macbeth to evaluate the validity of the factors.

Hong and White, 1995 is a specification test with series regression.

Fan and Yao, 2008 also has a chapter on generalized likelihood ratio statistic.

Apparently Horowitz and Spokoiny, 2001 is an important article, it's also mentioned in Li and Racine, 2007.

Here the econometric problem is to extend the GLR test to incorporate first-step estimation error.

I think some work need be done for the estimation of the covariance because there seem to be cross-sectional correlation. We can show asymptotic normality, see Li and Racine, 2007 chapter 2 and 18, should also check chapter 19 and 20. Then we follow

Hardle and Mammen, 1993 to see whether the linearity holds, which proposes a L^2 test and suggest bootstrap method. For that purpose, we need to find uniform convergence rate, see Mack and Silverman, 1982, which is based on Komlós, Major, and Tusnády, 1975, see Berkes, Liu, and Wu, 2014 for the extension to dependent data. Komlós, Major, and Tusnády, 1975 states a result due to Bartfai 1966, and later a simple proof for that is proved in ERD, 1970.

As pointed out by Fama and MacBeth, 1973, we should calculate that $Eg(\cdot)$ is a linear function rather then $g(\cdot)$ being linear.

Okui and Yanagi, 2019 provides a kernel density estimation method with the first-step estimation error. Dhaene and Jochmans, 2015 provides a way to correct for biases.

Fama and Macbeth mentioned that the estimated $\hat{\beta}_i$ tends to be above the true β_i for large β, I wonder why that's the case.

Cochrane, 2001 provides a way to put the two passes in one pass with GMM framework. Tao, 2020 might also be helpful.

$$R_{it} = h_t(\beta_i) + e_{it}$$

where we estimate β_i from the time series of returns. Then each period we have h_t is only driven by the cross-section size and the time series estimation error in β_i. We want to test that $h_t(b) = b * R_m t$ for some observed $R_m t$ or even some latent market factor so $b * U_t$, since then we would have $h_t(b)/h_s(b) = U_t/U_s$ which does not vary with b, which would be testable

In section 3.5 of Gagliardini, Ossola, and Olivier Scaillet, 2016, they propose a specification test and mentions another paper Kan, Robotti, and Shanken, 2013 which describes the asymptotic distribution of R^2 statistic and comparison between different asset pricing models' R^2.

Fan and Jiang, 2007 is a review article on GLR. It also mentions some potential questions in that area. I wonder if it's possible to combine Horowitz and Spokoiny, 2001 with GLR.

Escanciano, Pardo-Fernández, and Keilegom, 2018 seems to consider the case when there is stationary and mixing dependent data. I wonder if it applies to the case when we have cross-sectionally correlated data.

There are other things to think about: for example, local linear regression in the

settings of Andrews, 2005 and GLR test with dependent data.

2 Introduction and Model Set-up

This paper is about testing the validity of a linear factor asset pricing model using two-pass regression method.

Here we take the view that the β's of an asset is an important/fundamental property of the asset, so instead of considering where β are potentially nonlinear functions of conditioning variables, i.e., $\beta = \beta(z)$, or considering the time varying risk premium, which still specifies that the expected return is a linear function of β, we test whether the expected return is a nonlinear function of β.

Suppose there are k-dimenisonal observed candidate factors $F_t := (F_{1,t}, \dots, F_{k,t})$ for $t = 1, \dots, T$. A linear factor asset pricing model specifies the returns Y_{it} for asset/stock i at time t as

$$Y_{it} = \beta_i' F_t + \varepsilon_{it} \tag{2}$$

Assumption 1 (Model under H_0)**.** *In addition to the model in Equation 2, we make the following assumptions on the data generating process*

1. *β_i are independent across i and $\beta = \left[\beta_1, \dots, \beta_N\right]'$ is independent of F_t for all t.*
2. *$\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Nt})'$ are independent over time t. $E\varepsilon_t\varepsilon_t' = \Sigma$, there may be cross-sectional correlation. Assume that $\|\Sigma\| < \infty$.*
3. *F_t are stationary and $E(\varepsilon_t \mid \beta, F_t) = 0$.*

If we take expectation on both sides of Equation 2, we have the return-beta pricing equation:

$$E(Y_i \mid \beta_i) = \beta_i' \gamma \quad \text{where} \quad \gamma = E(F) \tag{3}$$

In order to test the model in Equation 2, we embed Equation 3 in a larger model:

$$E(Y_i \mid \beta_i) = g(\beta_i, s_i) \quad \text{or} \tag{4}$$

$$= g(\beta_i) + s_i' \gamma_s \tag{5}$$

where s_i is a vecotr of other candidate measure of risk.

The testable restrictions we derive from Equation 3 are,

1. $g(x, y) = g(x)$, that is no other risk measure will be priced. (Or $\gamma_s = 0$). β_i will fully explain the cross-sectional variation in expected excess returns.

2. $g(x) = x'\gamma_1$, the return is a linear function of the risk exposure.

3. $g(0) = 0$, there is no pricing error, which is the usual test of α.

3 Inference

Let's first assume there is no s_i. We write vectors in columns.

We construct the test statistic using the two-pass regression method. We calculate the residual estimated under the null specification as well as the nonparametric alternative and compare the difference.

There are two procedures commonly used:

1. the cross-sectional regression approach: where we first estimate the expected excess return with $\bar{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}$ and $\bar{F} = \frac{1}{T}\sum_{t=1}^{T} F_t$ and then run a single cross-sectional regression. Shanken, 1992a.

2. the Fama-MacBeth runs a cross-sectional regression for each time t, obtain the regression estimate $\hat{\gamma}_t$ for each t and the use $\hat{\gamma}_t$ as a sample for γ. This corrects for the cross-sectional correlation but not the error from first step estimation of β.

3.1 Estimation Residual Under the Null

First we estimate each stock's β's via the time-series regression:

$$Y_i = F\beta_i + \varepsilon_i \quad \text{for} \quad i = 1, \dots, N$$

where F is $T \times k$ and $Y_i : T \times 1$.

$$\hat{\beta}_i = \beta_i + \left(F'F\right)^{-1} F'\varepsilon_i$$

In the cross-sectional regression, under the null we will have the following regression model for each time t,

$$Y_{it} = \beta_i' \gamma + u_{it} \tag{6}$$

with

$$u_{it} = Y_{it} - \beta_i' \gamma_t = \varepsilon_{it} + \beta_i'(F_t - E(F))$$

which satisfies

$$E(u_{it} \mid \beta_i) = 0.$$

Let $\beta = (\beta_1, \dots, \beta_N)'$ which is $N \times k$. Equation 6 can be written in the matrix form as

$$Y_t = \beta \gamma_t + u_t \tag{7}$$

Suppose we replace β with $\hat{\beta}$ from first step and estimate the γ using cross-sectional OLS then

$$\hat{\gamma}_t = \left(\hat{\beta}'\hat{\beta}\right)^{-1} \hat{\beta}' Y_t = \left(\hat{\beta}'\hat{\beta}\right)^{-1} \hat{\beta}' [\beta F_t + \varepsilon_t]$$

Then

$$\begin{aligned} \hat{u}_{it} &= Y_{it} - \hat{\beta}_i' \hat{\gamma} \\ &= \beta_i' F_t + \varepsilon_{it} - \hat{\beta}_i' \hat{\gamma}_t \end{aligned}$$

Stack the $\hat{u}_{it}$ into $\hat{u}_t$ we have

$$\begin{aligned} \hat{u}_t &= \beta F_t + \varepsilon_t - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1} \hat{\beta}' [\beta F_t + \varepsilon_t] \\ &= \left(I - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1} \hat{\beta}'\right) [\beta F_t + \varepsilon_t] \\ &= \left(I - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1} \hat{\beta}'\right) \left[u_t - \left(\hat{\beta} - \beta\right) \gamma_t\right] \end{aligned}$$

Let $\bar{\varepsilon}_i = \frac{1}{T} \sum_t \varepsilon_{it}$ or ε_{it} corresponding to cross-sectional regression or Fama-MacBeth.

$\bar{\gamma} = \gamma + \bar{F} - E(F)$, we have

$$\hat{u}_i = \bar{\varepsilon}_i - \hat{\beta}_i \hat{A}_\beta \bar{\varepsilon} - \left\{ \left(\hat{\beta}_i - \beta_i \right) \bar{\gamma} - \hat{\beta}_i \hat{A}_\beta \left(\hat{\beta} - \beta \right) \bar{\gamma} \right\}$$
$$= \begin{cases} O_p(1) + O_p\left(\frac{1}{\sqrt{N}}\right) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right) & \text{if we don't do the time averaging first} \\ O_p(\frac{1}{\sqrt{T}}) + O_p\left(\frac{1}{\sqrt{NT}}\right) + O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right) & \text{if we do the time averaging first} \end{cases}$$

3.2 Estimation Residual Under the Alternative

Assume we have

$$m(x) := E(Y_i \mid \beta_i = x)$$

Then by similar procedure we have

$$\hat{m}(x) = \left(\hat{X}_D' \hat{W}_D \hat{X}_D \right)^{-1} \hat{X}_D' \hat{W}_D \bar{Y} = \left(\hat{X}_D' \hat{W}_D \hat{X}_D \right)^{-1} \hat{X}_D' \hat{W}_D \left[\bar{M} + \bar{\varepsilon} \right]$$

where $\hat{X}$ is the estimated betas and $\hat{W}$ is a diagonal matrix consisting of the K_h.

We embed the model in a nonparametric regression model and estimate with local linear method.

Follow the procedure in Li and Racine, 2007 or Hardle and Mammen, 1993, we test the specification based on the error term $\hat{u}_t = Y_t - \hat{\beta}\hat{\gamma}$, under H_0 we have:

$$E(uE(u \mid \beta = b)f(b)) = 0 \quad a.s.$$

Following Li and Racine, 2007, we construct a Nadaraya-Watson type kernel regression estimator. Let $K_{h,ij} = \prod_{i=1}^{k} k\left(\left(X_i - X_j\right)/h\right)$, then we try to use the relationship:

$$E[uE(u \mid \beta = b)f(b)] = 0 \quad a.s. - b \quad \text{under} \quad H_0$$

Our test statistic is based on a transformed and studentized version of the following statistic:

$$I = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j \neq i} \hat{u}_i \hat{u}_j K_{h,ij}$$

4 Empirical Studies

We collect data from CRSP, 1960-2019, all stocks with NaN are discarded, leaving with 304 stocks. Use window length of 100 days, 50 days overlapping. Calculate the beta in the first window and sort the stocks into 31 portfolios. Then calculate the betas in each window and the return in that period,

5 Further Questions

1. Test for nonlinearity in the beta: should we use sieve; should we use daily data of each stocks; the error-in-variable problem. Can we put it in the SDF framework.

2. Inner-product structure fits the real data, and so the arbitrage pricing theory implies linear pricing function in that space?

3. The eigenvectors seem to be stable, can it be a random thing?

4. Can we extract the signal from the time-varying betas.

5. About the exogeneity of the market portfolio, does it make sense to exclude large-value stocks so that the weights are small.

6. General question: is there a similarity between time series and the high-frequency literature.

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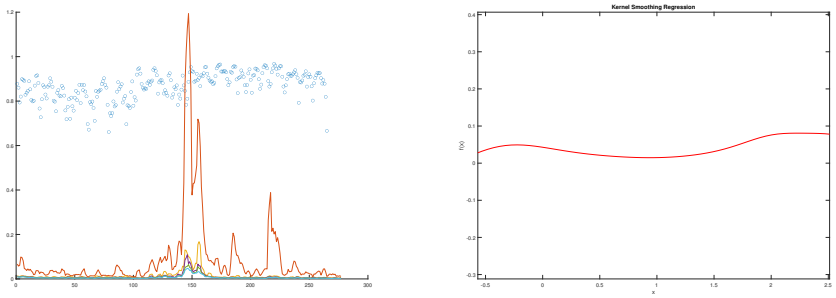

Figure 1: Kernel smoothing